

HOME-WORK 4

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Question 1 [20 Marks]

A 4 KiB cache can hold 4 KiB of data; however, caches are also required to store metadata such as tags and valid bits. We will see how block size affects the size of metadata needed for a direct-mapped cache. For all parts, assume that addresses are 64 bits, and we are dealing with words of 4 bytes.

a) Calculate the total number of bits required to implement a 128 KiB cache with 4 words per block. [5 Marks]

- block size = 4 words $\times$ 4 bytes/word = 16 bytes
- Number of blocks = $128 \times 1024 = 131,072$ bytes
which is,
 $131072/16 = 8192$ blocks
- Index bits = $\log_2(8192) = 13$ bits
- offset bits = $\log_2(16) = 4$ bits
- Tag bits = $64 - 13 - 4 = 47$ bits
- Line size = Data (4 words $\times$ 32 bits = 128 bits) + Tag (47 bits) + Valid bit (1 bit) = 176 bits
- Total cache size: $8192 \text{ blocks} \times 176 \text{ bits/block} = 1441792 \text{ bits}$

b) Now repeat the calculation in part a with 8, 16, 32 and 64 words per block. [10 Marks]

Block size	8 words/block	16 words/ block	32 words/block	64 words/block
# of cache blocks	4096	2048	1024	512
Bits for Index	12	11	10	9
Bits for Tag	47	47	47	47
Line Size in bits	304	560	1072	2096
Total Size in bits	1245184	1146880	1097728	1073152

c) Find the ratio of total cache size to the amount of data (i.e. data size) for all the cases in part 'a' and 'b'. [5 Marks]

- We will divide Total size of every part with: 1048576 bits,

Block Size	Ratio
4 words/blocks	1.375
8 words/block	1.187
16 words/block	1.094
32 words/block	1.047
64 words/block	1.023

$$\therefore 128 \text{ kib} \times 1024 \text{ bytes/kib} \times 8 \text{ bits/byte} = 1048576 \text{ bits}$$

Question 3 [10 Marks]

Cache block size (B) can affect both miss rate and miss latency. Assuming a machine with a base CPI of 1, and an average of 1.35 references (both instruction and data) per instruction, find the block size that minimizes the total miss latency given the following miss rates for various block sizes. Cache block size is assumed to be in bytes.

8: 10%	16: 8%	32: 7%	64: 5%	128: 3%
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a) What is the optimal block size for a miss latency of $20 \times B$ cycles? [5 Marks]

- Miss latency = $20 \times B$ cycles:

$$B=8: \text{latency} = 160, \text{penalty} = 0.1 \times 160 = 16.0$$

$$B=16: \text{latency} = 320, \text{penalty} = 0.08 \times 320 = 25.6$$

$$B=32: \text{latency} = 640, \text{penalty} = 0.07 \times 640 = 44.8$$

$$B=64: \text{latency} = 1280, \text{penalty} = 0.05 \times 1280 = 64.0$$

$$B=128: \text{latency} = 2560, \text{penalty} = 0.03 \times 2560 = 76.8$$

- as block size 8 minimizes miss penalty so it's the optimal size.

b) What is the optimal block size for a miss latency of $24 + B$ cycles? [5 Marks]

$$\begin{aligned}
 B=8 : \text{ Latency} &= 32, \text{ Penalty} = 0.1 \times 32 = 3.2 \\
 B=16 : \text{ Latency} &= 40, \text{ Penalty} = 0.08 \times 40 = 3.2 \\
 B=32 : \text{ Latency} &= 56, \text{ Penalty} = 0.07 \times 56 = 3.92 \\
 B=64 : \text{ Latency} &= 88, \text{ Penalty} = 0.05 \times 88 = 4.4 \\
 B=128 : \text{ Latency} &= 152, \text{ Penalty} = 0.03 \times 152 = 4.56
 \end{aligned}$$

- Block size of 8 bytes or 16 bytes are optimal sizes as they yield minimum Miss penalty.

Question 4 [10 Marks]

We are given 4 arrays of size 6. Each element of an array is of 32 bytes i.e., one word. Following is the data stored in the array:

A = (27, 11, 23, 10, 28, 12)

B = (20, 35, 33, 48, 10, 32)

C = (40, 11, 9, 42, 49, 20)

D = (11, 31, 17, 37, 13, 2)

Data to be Inserted	Hit/Miss	Cache Index							
		0	1	2	3	4	5	6	7
A[0]	M	A[0]							
A[1]	M	A[0]	A[1]						
A[2]	M	A[0]	A[1]	A[2]					
A[1]	H	A[0]	A[1]	A[2]					
A[5]	M	A[0]	A[1]	A[2]			A[5]		
B[5]	M	A[0]	A[1]	A[2]			B[5]		
B[4]	M	A[0]	A[1]	A[2]		B[4]	B[5]		
B[3]	M	A[0]	A[1]	A[2]	B[3]	B[4]	B[5]		
B[3]	H	A[0]	A[1]	A[2]	B[3]	B[4]	B[5]		
B[4]	H	A[0]	A[1]	A[2]	B[3]	B[4]	B[5]		
D[1]	M	A[0]	D[1]	A[2]	B[3]	B[4]	B[5]		
D[2]	M	A[0]	D[1]	D[2]	B[3]	B[4]	B[5]		
D[3]	M	A[0]	D[1]	D[2]	D[3]	B[4]	B[5]		
D[4]	M	A[0]	D[1]	D[2]	D[3]	D[4]	B[5]		
C[3]	M	A[0]	D[1]	D[2]	C[3]	D[4]	B[5]		
C[2]	M	A[0]	D[1]	C[2]	C[3]	D[4]	B[5]		
C[4]	M	A[0]	D[1]	C[2]	C[3]	C[4]	B[5]		
C[2]	H	A[0]	D[1]	C[2]	C[3]	C[4]	B[5]		

- Hit ratio = 4 hits / 18 accesses = 22.2%
- Miss ratio = 14 misses / 18 accesses = 77.8%

Question 5 [10 Marks]

Repeat the same task as in Question 4 but now the cache has 4 lines and the block size has increased to two words. The first insertion is done again so that you may get the idea.

Data to be Inserted	Hit/Miss	Cache Index							
		0		1		2		3	
A[0]	M	A[0]	A[1]						
A[1]	H	A[0]	A[1]						
A[2]	M	A[0]	A[1]	A[2]	A[3]				
A[1]	H	A[0]	A[1]	A[2]	A[3]				
A[5]	M	A[0]	A[1]	A[2]	A[3]	A[4]	A[5]		
B[5]	M	A[0]	A[1]	A[2]	A[3]	B[4]	B[5]		
B[4]	H	A[0]	A[1]	A[2]	A[3]	B[4]	B[5]		
B[3]	M	A[0]	A[1]	B[2]	B[3]	B[4]	B[5]		
B[3]	H	A[0]	A[1]	B[2]	B[3]	B[4]	B[5]		
B[4]	H	A[0]	A[1]	B[2]	B[3]	B[4]	B[5]		
D[1]	M	D[0]	D[1]	B[2]	B[3]	B[4]	B[5]		
D[2]	M	D[0]	D[1]	D[2]	D[3]	B[4]	B[5]		
D[3]	H	D[0]	D[1]	D[2]	D[3]	B[4]	B[5]		
D[4]	M	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]		
C[3]	M	D[0]	D[1]	C[2]	C[3]	D[4]	D[5]		
C[2]	H	D[0]	D[1]	C[2]	C[3]	D[4]	D[5]		
C[4]	M	D[0]	D[1]	C[2]	C[3]	C[4]	C[5]		
C[2]	H	D[0]	D[1]	C[2]	C[3]	C[4]	C[5]		

What is the Hit Ratio and the Miss Ratio in this case? Is it better than the previous? Does loading the whole array into the cache help us in accessing the elements fast?

- Hit ratio = $8/18 = 44.4\%$
- Miss ratio = $10/18 = 55.6\%$
- Yes, increasing the block size increased spatial locality, which dropped the miss rate from 77.8% to 55.6%.
- Loading a full array at a time would eliminate conflict misses that occur when elements of different arrays map to the same cache line.